

SCX $\hat{S}$

Operational Definition of the Potential Operator $\hat{S}$ in SCX Social Inference

SCX

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Abstract

SCX $\hat{S}$ $\hat{S}$ GDP / / / / / $\sum_m g_m = 0$ SCX

We define the operational form of the potential operator $\hat{S}$ for the social-inference module of the SCX egalitarian framework. $\hat{S}$ maps social entities to scalar potential values across three dimensions: **national potential** (per-capita GDP / Gini coefficient / life expectancy), **wealth potential** (net-asset quantile / income quantile), and **cognitive potential** (years of education / citation count / abstract-thinking test). Each operator satisfies four conditions: quantifiability, auditability, the gauge-fixing condition $\sum_m g_m = 0$, and cross-domain comparability. We formalize these definitions rigorously within the SCX framework and provide audit protocols and empirical calibration paths.

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1 Introduction and Framework Background

1.1 SCX Social Inference in SCX Egalitarianism

SCX $\mathcal{F} : \mathcal{U} \rightarrow \mathbb{R}$

One of the core insights of the SCX framework is that inequality in social systems is not merely a statistical phenomenon, but a **field-theoretic problem**. Differences among social entities (individuals, households, regions, nations) can be modeled as local gradient effects of a social scalar field $\mathcal{F} : \mathcal{U} \rightarrow \mathbb{R}$, where $\mathcal{U}$ is the universe of social entities.

SCX

(1) **Cercis** $S = Q + \eta N$ Q N η -

(2) $\hat{S}$

In the SCX formal apparatus, we distinguish two core operators:

- (1) **Cercis Score operator** S : the audit tool measuring data quality, defined over Situs encoding space, satisfying $S = Q + \eta N$, where Q is the quality component, N the noise component, and η the noise-hardness coupling parameter.
- (2) **Potential operator** $\hat{S}$ (the subject of this paper): the operational definition measuring a social entity's *relative position* or *potential* in a given dimension, constituting the foundational input for social inference.

1.2 Four Design Constraints

$\hat{S}$

Any valid potential operator $\hat{S}$ must satisfy four axiomatic conditions:

Axiom 1.1 (Quantifiability). $e \in \mathcal{U} \hat{S}(e) \in \mathbb{R}$

For any social entity $e \in \mathcal{U}$, $\hat{S}(e)$ is a well-defined real number, and there exists a finite algorithm that outputs an estimate of $\hat{S}(e)$ in finite time given observational data about e .

Axiom 1.2 (Auditability). $\hat{S}$ audit trail $T(e) = \{d_1, \dots, d_k\}$ A $T(e)$ $\hat{S}(e)$ SCX T5

Every concrete computation instance of $\hat{S}$ must carry an **audit trail** $T(e) = \{d_1, \dots, d_k\}$ such that any independent auditor A can verify the correctness of $\hat{S}(e)$ using $T(e)$, and the sample complexity of the verification process satisfies the SCX T5 active-learning bounds.

Axiom 1.3 (Gauge Fixing). $\mathcal{U} \hat{S}$

$$\sum_{e \in \mathcal{U}} g_e(\hat{S}) = 0 \quad (1)$$

$$g_e(\hat{S}) = \hat{S}(e) - \bar{\hat{S}} = |\mathcal{U}|^{-1} \sum_e \hat{S}(e) - \hat{S}(e) = 0$$

For all entities in domain $\mathcal{U}$, the value of $\hat{S}$ satisfies the gauge-fixing condition as in Eq. (1), where $g_e(\hat{S})$ is the gauge-weight function for entity e . Under the simplest uniform gauge, $g_e = \hat{S}(e) - \bar{\hat{S}}$, where $\bar{\hat{S}} = |\mathcal{U}|^{-1} \sum_e \hat{S}(e)$, i.e., $\sum_e \hat{S}(e) = 0$.

Axiom 1.4 (Cross-Domain Comparability). $e_1, e_2 \in \mathcal{U}_1, \mathcal{U}_2$ $\tau_{12} : \mathbb{R} \rightarrow \mathbb{R}$ $\hat{S}(e_1) \sim \tau_{12}(\hat{S}(e_2))$ $\tau_{12} \circ \tau_{23} = \tau_{13}$

If two entities e_1, e_2 belong to distinct social domains $\mathcal{U}_1, \mathcal{U}_2$, there exists a gauge transformation $\tau_{12} : \mathbb{R} \rightarrow \mathbb{R}$ such that $\hat{S}(e_1)$ and $\tau_{12}(\hat{S}(e_2))$ are comparable, and τ_{12} satisfies transitivity: $\tau_{12} \circ \tau_{23} = \tau_{13}$.

1.3 Physical and Social Meaning of Gauge Fixing

$$\sum_m g_m = 0$$

The gauge-fixing condition $\sum_m g_m = 0$ originates from standard techniques in gauge field theory. In the social-field-theoretic context, its meaning is:

1. **Principle of Relativity** Social potential is a relative quantity, not an absolute one. Just as only potential differences have physical meaning, social potential only has meaning when compared against a reference group.
2. **Zero-Sum Calibration** Gauge fixing ensures the global mean of social potential is zero. This means potential measures deviation from the global mean, not some absolute endowment.
3. **Elimination of Redundant Degrees of Freedom** GDP The absolute value of social potential depends on an arbitrarily chosen unit system (e.g., whether GDP is measured in USD or CNY). Gauge fixing eliminates this redundancy, rendering potential values invariant under rescaling.
4. **Group Structure** $\mathcal{G} \cong (\mathbb{R}, +)$ $g, g', c \in \mathbb{R}$ $g' = g + c$ Gauge transformations form an additive group $\mathcal{G} \cong (\mathbb{R}, +)$. Any two gauge choices g and g' are related by a translation $c \in \mathbb{R}$: $g' = g + c$. Gauge fixing is equivalent to quotienting by this group action.

2 S General Form of $\hat{S}$

2.1 Formal Definition

Definition 2.1 ($\hat{S}$). $\mathcal{U} \times \mathcal{M} \rightarrow \mathbb{R}$

$$\hat{S} : \mathcal{U} \times \mathcal{M} \rightarrow \mathbb{R} \quad (2)$$

$\mathcal{M} = \{\text{nat}, \text{wealth}, \text{cog}, \dots\}$ $e \in \mathcal{U}$ $m \in \mathcal{M}$

$$\hat{S}(e; m) = \Phi_m(x_1^{(m)}(e), x_2^{(m)}(e), \dots, x_{k_m}^{(m)}(e)) \quad (3)$$

$$\{x_i^{(m)}(e)\}_{i=1}^{k_m} : \mathbb{R}^{k_m} \rightarrow \mathbb{R}$$

Let $\mathcal{U}$ be the universe of social entities and m a measurement-dimension index. The potential operator $\hat{S}$ is defined as a functional as in Eq. (3), where $\mathcal{M} = \{\text{nat}, \text{wealth}, \text{cog}, \dots\}$ is the set of measurement dimensions. For entity $e \in \mathcal{U}$ and dimension $m \in \mathcal{M}$, $\{x_i^{(m)}(e)\}_{i=1}^{k_m}$ are the k_m observable features of the entity under dimension m , and $\Phi_m : \mathbb{R}^{k_m} \rightarrow \mathbb{R}$ is the dimension-specific aggregation function.

2.2 Design Principles for Φ_m

Φ_m

Each Φ_m must satisfy the following design principles:

DP1: Monotonicity $\partial \Phi_m / \partial x_i \geq 0$ $\partial \Phi_m / \partial x_j \leq 0$ For each positive feature x_i , $\partial \Phi_m / \partial x_i \geq 0$. For each negative feature x_j , $\partial \Phi_m / \partial x_j \leq 0$.

DP2: Homogeneity $\Phi_m(\lambda x_1, \dots, \lambda x_{k_m}) = \lambda \Phi_m(x_1, \dots, x_{k_m})$ $\lambda > 0$ Φ_m is homogeneous of degree 1. This ensures covariance of potential values under unit changes of features.

DP3: Additive Separability $\Phi_m(x_1, \dots, x_{k_m}) = \sum_{i=1}^{k_m} w_i \cdot f_i(x_i)$ $w_i \geq 0$ $\sum_i w_i = 1$ f_i

The aggregation is additively separable with weights $w_i \geq 0$, $\sum_i w_i = 1$, and feature normalization functions f_i . This ensures decomposable verifiability of the audit trail.

DP4: Robustness $\Phi_m \sup_x |IF(x; \Phi_m, F)| < \infty$ IF Hampel Φ_m is insensitive to outliers, satisfying bounded influence: $\sup_x |IF(x; \Phi_m, F)| < \infty$, where IF is the Hampel influence function.

2.3 Normalization Protocol and Gauge Fixing

$$m \tilde{S}_m(e) = \Phi_m(x_1^{(m)}(e), \dots, x_{k_m}^{(m)}(e))$$

Given raw aggregated values $\tilde{S}_m(e)$, the gauge-fixed potential operator is:

$$\hat{S}(e; m) = \frac{\tilde{S}_m(e) - \mu_m}{\sigma_m} \quad (4)$$

$$\mu_m = |\mathcal{U}|^{-1} \sum_{e \in \mathcal{U}} \tilde{S}_m(e) \quad \sigma_m = \sqrt{|\mathcal{U}|^{-1} \sum_{e \in \mathcal{U}} (\tilde{S}_m(e) - \mu_m)^2}$$

where μ_m is the global mean and σ_m the global standard deviation.

$$\sum_{e \in \mathcal{U}} \hat{S}(e; m) = 0, \quad \frac{1}{|\mathcal{U}|} \sum_{e \in \mathcal{U}} [\hat{S}(e; m)]^2 = 1 \quad (5)$$

$$\hat{S}(e; m) \sim N(0, 1) \quad \tilde{S}_m \quad (5) \quad 1.3 \quad g_e(\hat{S}) = \hat{S}(e; m)$$

This normalization automatically satisfies Eq. (5), which is precisely the uniform-gauge special case of Axiom 1.3.

3 $\hat{S}_{\text{nat}}$ National Potential Operator

3.1 Conceptual Definition

$$\hat{S}_{\text{nat}}$$

National potential $\hat{S}_{\text{nat}}$ measures the structural potential of a sovereign state (or equivalent political entity) to provide basic welfare conditions for its citizens. It integrates three fundamental dimensions:

- (1) **Economic Output** GDP PPP GDP per capita (PPP), reflecting material resource availability.
- (2) **Distributional Fairness** Gini coefficient, reflecting the equality of resource distribution (negative indicator).
- (3) **Life Quality** Life expectancy at birth, reflecting the combined outcome of health infrastructure and nutritional security.

Remark 3.1. Sen capability approach GDP Gini —

These three dimensions conceptually cover the core elements of Sen's capability approach: material resources (GDP), resource distribution (Gini), and basic functionings (life expectancy). Together they measure a nation's potential — the capability space in which citizens can achieve valuable functionings, rather than the achievements themselves.

3.2 Observable Variables and Data Sources

1.2 All data sources are from publicly accessible international databases, satisfying the audit requirement of Axiom 1.2.

Table 1: Observable Variables for National Potential

Variable	Symbol	Source		
GDP (PPP, 2017 int\$)	x_1	World Bank WDI / IMF WEO	Annual	+
	x_2	World Bank / SWIID	2–5	–
	x_3	WHO / UN Population Division	Annual	+

3.3 Aggregation Function Construction

The raw national potential aggregation is defined as a log-weighted average of three normalized features:

$$\tilde{S}_{\text{nat}}(e) = w_1 \cdot \log(x_1(e)) + w_2 \cdot \log(1 - x_2(e)) + w_3 \cdot \log(x_3(e)) \quad (6)$$

$$w_1, w_2, w_3 \geq 0 \quad \sum w_i = 1$$

where weights satisfy $w_1, w_2, w_3 \geq 0$ and $\sum w_i = 1$.

Rationale for the log transform:

1. GDP

GDP and life expectancy exhibit log-normal distributional characteristics; the log transform brings the distribution closer to normality.

2. $\hat{S}_{\text{nat}}(e_1) - \hat{S}_{\text{nat}}(e_2) \approx \log(\text{ratio})$ The log transform converts multiplicative differences to additive ones, giving potential differences a ratio interpretation.

3. $(1 - x_2(e)) \rightarrow x_2 \rightarrow 1$ $\log(1 - x_2) \rightarrow -\infty$ The complement $(1 - x_2)$ ensures high Gini (more unequal) $\rightarrow$ low potential, and as $x_2 \rightarrow 1$, $\log(1 - x_2) \rightarrow -\infty$, imposing automatic penalization.

3.4 Gauge Fixing and Final Form

$$\mu_{\text{nat}} = \frac{1}{N} \sum_{e=1}^N \tilde{S}_{\text{nat}}(e) \quad (7)$$

$$\sigma_{\text{nat}} = \sqrt{\frac{1}{N} \sum_{e=1}^N (\tilde{S}_{\text{nat}}(e) - \mu_{\text{nat}})^2} \quad (8)$$

$$\boxed{\hat{S}_{\text{nat}}(e) = \frac{\tilde{S}_{\text{nat}}(e) - \mu_{\text{nat}}}{\sigma_{\text{nat}}}} \quad (9)$$

$\sum_e \hat{S}_{\text{nat}}(e) = 0$ —

Automatically satisfies $\sum_e \hat{S}_{\text{nat}}(e) = 0$ — the global mean of national potential is calibrated to zero. Positive values indicate above-average national potential; negative values indicate below-average.

3.5 Empirical Calibration of Weights

$\mathbf{w} = (w_1, w_2, w_3)$

The weight vector $\mathbf{w} = (w_1, w_2, w_3)$ can be determined through two complementary methods:

1: PCA $\mathbf{Z}_{N \times 3}$

$$\mathbf{w}_{\text{PCA}} = \frac{|\mathbf{v}_1|}{\|\mathbf{v}_1\|_1} \quad (10)$$

$\mathbf{v}_1 \mathbf{Z}^\top \mathbf{Z}$

Take the loading vector of the first principal component; this maximizes the variance-explained ratio of potential values.

2: Delphi Expert Consensus $K \geq 5$ $\mathbf{w}^{(k)}$ $\mathbf{w}_{\text{Delphi}} = \text{median}_k(\mathbf{w}^{(k)})$ AC-Theorem $K_{\min}$
Organize $K \geq 5$ development economists; take the median weight vector, satisfying the AC-Theorem independent-expert lower bound.

3: Fusion $\mathbf{w} = \alpha \mathbf{w}_{\text{PCA}} + (1 - \alpha) \mathbf{w}_{\text{Delphi}}$ $\alpha \in [0, 1]$

Remark 3.2. $\mathbf{w} = (1/3, 1/3, 1/3)$

Default recommended weights are equal weights $\mathbf{w} = (1/3, 1/3, 1/3)$ as an uninformative prior, unless strong empirical evidence supports deviation from equality. This minimizes dependence on specific value judgments.

3.6 Audit Protocol

1

Algorithm 1 $\mathcal{A}_{\text{nat}}$

```

1: Input  $\mathcal{L} = \{e_1, \dots, e_N\}$   $B$ 
2: Output  $\mathbf{S} = (\hat{\mathcal{S}}_{\text{nat}}(e_1), \dots, \hat{\mathcal{S}}_{\text{nat}}(e_N))$ 
3: 1 Data Provenance
4: for  $e \in \mathcal{L}$  do
5:   WDI/WEO  $x_1(e)$  API
6:   SWIID  $x_2(e)$  /
7:   WHO  $x_3(e)$ 
8: end for
9: 2 Consistency Check
10:  $(e_i, e_j) \leq 3$ 
11:  $> 3$  Temporal Comparability Warning
12: 3 Aggregation
13:  $\tilde{\mathcal{S}}_{\text{nat}}(e) \leftarrow \sum_{i=1}^3 w_i \cdot f_i(x_i(e))$ 
14: 4 Gauge Fixing
15:  $\mu \leftarrow N^{-1} \sum_e \tilde{\mathcal{S}}(e)$   $\sigma \leftarrow \sqrt{N^{-1} \sum_e (\tilde{\mathcal{S}}(e) - \mu)^2}$ 
16:  $\left| \sum_e \hat{\mathcal{S}}_{\text{nat}}(e) \right| < \epsilon_{\text{machine}}$ 
17: 5 Anomaly Detection
18:  $|\hat{\mathcal{S}}_{\text{nat}}(e)| > 3$ 
19: 6 Publish Audit Trail
20:  $\{(e, x_1, x_2, x_3, \tilde{\mathcal{S}}, \hat{\mathcal{S}}_{\text{nat}}, \text{timestamp}, \text{source\_version})\}_{e \in \mathcal{L}}$ 
21: SHA-256
```

1 $\hat{\mathcal{S}}_{\text{nat}}(e)$ 1.2 This protocol guarantees that any independent auditor can fully reproduce the values of $\hat{\mathcal{S}}_{\text{nat}}(e)$ using only public data and Algorithm 1, satisfying Axiom 1.2.

3.7 Illustrative Quantification

2023

Table 2: Example National Potential Computation

	GDP/cap	Gini		$\hat{S}_{\text{nat}}$	$\hat{S}_{\text{nat}}$	
A	\$65,000	0.30	83.5	1.523	+1.42	1
B	\$12,000	0.45	75.0	-0.087	-0.18	2
C	\$1,200	0.55	62.0	-1.436	-1.24	3
	—	—	—	0.000	0.000	—

$$\hat{S}_{\text{nat}}(A) = +1.42 \quad \hat{S}_{\text{nat}}(C) = -1.24 \quad \text{C1.24 2.66}$$

$\hat{S}_{\text{nat}}(A) = +1.42$ indicates Country A's national potential is 1.42 standard deviations above the global mean. $\hat{S}_{\text{nat}}(C) = -1.24$ indicates Country C's is 1.24 below. The difference of 2.66 standard deviations measures the extreme span of global national potential.

4 $\hat{S}_{\text{wealth}}$ Wealth Potential Operator

4.1 Conceptual Definition

$\hat{S}_{\text{wealth}}$ - ———

Wealth potential $\hat{S}_{\text{wealth}}$ measures the relative economic position of an individual or household in asset-income space. Unlike absolute wealth, wealth potential emphasizes the position and potential in the distribution — the spatial mapping of an individual's ability to use economic resources to achieve goals.

- (1) **Net Asset Stock** Total assets minus total liabilities, reflecting accumulated economic safety cushion.
- (2) **Income Flow** + Annual disposable income (post-tax + transfers), reflecting current economic vitality.

Remark 4.1. Distinguishing stock and flow is central to wealth-potential design. A retiree may have high net assets (stock) but low income (flow); a young professional may have high income but low net assets. Using either alone systematically under- or over-estimates economic potential for specific populations. Integrating both yields a more complete picture.

4.2 Quantile-Based Definition

The core innovation of wealth potential is the use of **quantiles** rather than raw values. The reasons are:

1. **Distributional Robustness** Pareto $U(0,1)$ Wealth and income exhibit extreme Pareto tail behavior. The quantile transform maps any distribution to $U(0,1)$, eliminating outlier distortion.
2. **Cross-Domain Comparability** $[0,1]$ / Quantiles are naturally normalized to $[0,1]$, making wealth potentials directly comparable across countries/periods (within their respective reference distributions).

3. Privacy-Friendly

Publishing quantiles rather than raw values reduces re-identification risk, satisfying basic differential privacy requirements.

4.3 Aggregation Function Construction

$$Q_{\text{net}}(e) \in [0, 1] \quad Q_{\text{inc}}(e)$$

Let $Q_{\text{net}}(e)$ be the net-asset quantile of entity e in the reference population (range $[0, 1]$), and $Q_{\text{inc}}(e)$ the disposable-income quantile. The raw wealth potential is:

$$\tilde{S}_{\text{wealth}}(e) = v_1 \cdot \Phi^{-1}(Q_{\text{net}}(e)) + v_2 \cdot \Phi^{-1}(Q_{\text{inc}}(e)) \quad (11)$$

$$\Phi^{-1} \text{ probit } [0, 1] \rightarrow \mathbb{R} \quad v_1, v_2 \geq 0 \quad v_1 + v_2 = 1$$

where Φ^{-1} is the quantile function (probit transform) of the standard normal distribution, mapping $[0, 1]$ to $\mathbb{R}$. Weights $v_1, v_2 \geq 0$, $v_1 + v_2 = 1$.

Probit Rationale for the probit transform:

1. $Q \sim U(0, 1) \rightarrow \Phi^{-1}(Q) \sim N(0, 1)$ copula After the probit transform, variables follow $N(0, 1)$, allowing simple additive combination without complex copula modeling.
2. The double-tailed nature of the normal distribution naturally reflects extreme wealth (upper tail) and extreme poverty (lower tail).

4.4 Gauge Fixing and Final Form

$$\hat{S}_{\text{wealth}}(e) = \frac{\tilde{S}_{\text{wealth}}(e) - \mu_{\text{wealth}}}{\sigma_{\text{wealth}}} \quad (12)$$

$$\mu_{\text{wealth}} = N^{-1} \sum_e \tilde{S}_{\text{wealth}}(e) \quad \sigma_{\text{wealth}} = \sqrt{N^{-1} \sum_e (\tilde{S}_{\text{wealth}}(e) - \mu_{\text{wealth}})^2}$$

$\sum_e \hat{S}_{\text{wealth}}(e) = 0$ The gauge-fixing condition $\sum_e \hat{S}_{\text{wealth}}(e) = 0$ is automatically satisfied.

4.5 Audit Protocol

Wealth potential data sources depend on the domain:

Table 3: Data Sources for Wealth Potential

Domain	Data Source	Audit Method
China	CFPS / CHFS ()	+
USA	SCF / PSID (Survey of Consumer Finances)	Fed + IRS
EU	HFCS (Household Finance and Consumption Survey)	ECB
Global	WID.world / Credit Suisse Global Wealth Report	

$\{e, Q_{\text{net}}, Q_{\text{inc}}, \tilde{S}, \hat{S}_{\text{wealth}}, \text{survey_wave}, \text{imputation_flag}\}$

The audit trail includes entity ID, quantiles, raw and gauge-fixed potentials, survey wave, and imputation flags. Auditors can verify quantile values through independent surveys or public microdata sampling.

Table 4: Example Wealth Potential Computation

	Q_{net}	Q_{inc}	$\tilde{S}$	$\hat{S}_{\text{wealth}}$	
Top 1%	0.99	0.97	+3.02	+2.85	99.8%
Median Middle	0.50	0.50	0.00	0.00	50.0%
Bottom 10%	0.05	0.08	-2.65	-2.50	0.6%

4.6 Illustrative Quantification

2022CHFS

$$\hat{S}_{\text{wealth}} \hat{S}_{\text{wealth}} = +2.85 \ 2.85$$

The median-zero value of $\hat{S}_{\text{wealth}}$ is a direct consequence of gauge fixing: the mean wealth potential of the reference population is identically zero. $\hat{S}_{\text{wealth}} = +2.85$ means the household’s economic potential is 2.85 standard deviations above the population median. This provides an economic-position metric robust to extreme values.

5 $\hat{S}_{\text{cog}}$ Cognitive Potential Operator

5.1 Conceptual Definition

$\hat{S}_{\text{cog}}$ latent variable

Cognitive potential $\hat{S}_{\text{cog}}$ measures an individual’s potential for processing abstract information, generating knowledge, and solving complex problems. This is the most challenging of the three potential dimensions, as cognitive ability is a latent variable that cannot be directly observed and must be measured indirectly through proxy variables.

- (1) **Education Years** Total years of formal education (including tertiary), reflecting institutionalized knowledge acquisition.
- (2) **Research Output-**
Field-and-year-normalized citation count, reflecting the quality and impact of knowledge production.
- (3) **Abstract Reasoning**Raven
Standardized abstract-reasoning test score (e.g., Raven’s Progressive Matrices), reflecting fluid intelligence.

Remark 5.1. Cattell-Horn-Carroll $\rightarrow G_c \rightarrow G_f \rightarrow G_{kn}$

These three proxies correspond to three strata in the Cattell-Horn-Carroll theory: education $\rightarrow$ crystallized intelligence (G_c), abstract reasoning $\rightarrow$ fluid intelligence (G_f), citations $\rightarrow$ domain-specific knowledge (G_{kn}). Together they form a multidimensional proxy for cognitive potential, but cannot claim to exhaust all aspects of cognitive ability.

5.2 Aggregation Function Construction

$\approx < 1\%$

Since citation data applies only to the academic research population ($\approx < 1\%$ of total population), cognitive potential must handle **missing-variable** issues. We define two versions:

A — Academic Population x_3

$$\tilde{S}_{\text{cog}}^{(A)}(e) = u_1 \cdot z_{\text{edu}}(e) + u_2 \cdot z_{\text{cite}}(e) + u_3 \cdot z_{\text{reason}}(e) \quad (13)$$

B — General Population x_3

$$\tilde{S}_{\text{cog}}^{(B)}(e) = u'_1 \cdot z_{\text{edu}}(e) + u'_3 \cdot z_{\text{reason}}(e) \quad (14)$$

$$z_{\text{var}}(e) = (x_{\text{var}}(e) - \bar{x}_{\text{var}}) / s_{\text{var}} \quad \text{z-score } u_i, u'_i$$

where $z_{\text{var}}(e)$ are the z-score normalized values of each variable, and u_i, u'_i are renormalized weights.

5.3 - Field-Year Normalization of Citations

Paper citations have strong field dependence and time dependence. Raw citation counts would produce severe systematic bias.

Relative Citation Rate, RCR

$$z_{\text{cite}}(e) = \frac{1}{|P(e)|} \sum_{p \in P(e)} \frac{c(p) - \mu(\text{field}(p), \text{year}(p))}{\sigma(\text{field}(p), \text{year}(p))} \quad (15)$$

$$P(e) \quad e \quad c(p) \quad p \quad \mu(f, y) \quad \sigma(f, y) \quad f \quad y$$

where $P(e)$ is the set of papers by entity e , $c(p)$ is the total citation count, and $\mu(f, y), \sigma(f, y)$ are the mean and standard deviation of citations in field f , publication year y . This normalization makes citation impact comparable across fields.

Remark 5.2. RCR $f \ y$ RCR01 RCRA key advantage of RCR: within the reference set of field f , year y , the mean of RCR is exactly 0 and its standard deviation is 1. This makes RCR naturally satisfy gauge-fixing within that reference set.

5.4 Gauge Fixing and Final Form

$$\hat{S}_{\text{cog}}(e) = \frac{\tilde{S}_{\text{cog}}(e) - \mu_{\text{cog}}}{\sigma_{\text{cog}}} \quad (16)$$

$$\mu_{\text{cog}}, \sigma_{\text{cog}} \quad \text{A B/}$$

$\mu_{\text{cog}}, \sigma_{\text{cog}}$ are computed based on the appropriate reference population (version A: global academic researchers; version B: national/regional adult population).

5.5 Audit Protocol

Auditing cognitive potential faces unique challenges: citation data is subject to gaming risks (citation rings, self-citation), and abstract-reasoning tests are subject to practice effects and test-condition variance.

The audit protocol includes the following anti-gaming measures:

1. **Self-Citation Removal** $c(p)$ OpenAlex/Semantic ScholarID Exclude all self-citations when computing $c(p)$, using disambiguated author IDs from OpenAlex/Semantic Scholar.

2. **Citation-Cartel Detection** Jaccard 3”” CitationCartelWarning *Compute inter-author citation Jaccard similarity matrix; if mutual citation rate within an author group exceeds 3 standard deviations, flag as ”Citation Cartel Warning.”*
3. **Test-Condition Calibration**
Abstract-reasoning scores must carry test-condition metadata (time limit, language, proctored, environment). Calibrate condition bias using multi-level models (random effects: test center).
4. **Repeated-Measures Correction** $s_{\text{adj}} = s_{\text{raw}} - \beta \cdot \log(k)$ $k \beta$ *If an individual takes the test multiple times, use only the first score (or apply decay correction).*

5.6 Illustrative Quantification

Table 5: A Example Cognitive Potential Computation

		RCR		$\hat{\mathcal{S}}_{\text{cog}}$	
Elite Researcher	22 (PhD+Postdoc)	+2.1	135 (99%ile)	+2.45	99.3%
Mid-Career	20 (PhD)	+0.3	115 (84%ile)	+0.22	58.7%
Early-Career	18 (MSc)	-0.5	110 (75%ile)	-0.61	27.1%

$$\hat{\mathcal{S}}_{\text{cog}} = +2.45 \text{ } 0.7\% \text{ RCR}$$

$\hat{\mathcal{S}}_{\text{cog}} = +2.45$ indicates this researcher’s cognitive potential is in the top 0.7% of the global academic population. Note: this metric reflects potential (the capability space based on education and cognitive tests), not achievement per se (though RCR already partially captures the achievement dimension).

6 Unified Theory and Inter-Operator Relations

6.1 Unified Form of the Three Operators

$$\hat{\mathcal{S}}_{\text{nat}}, \hat{\mathcal{S}}_{\text{wealth}}, \hat{\mathcal{S}}_{\text{cog}}$$

The three potential operators $\hat{\mathcal{S}}_{\text{nat}}, \hat{\mathcal{S}}_{\text{wealth}}, \hat{\mathcal{S}}_{\text{cog}}$ share a unified structure:

$$\hat{\mathcal{S}}_m(e) = \frac{1}{\sigma_m} \left(\sum_{i=1}^{k_m} w_i^{(m)} \cdot \psi_i^{(m)}(x_i^{(m)}(e)) - \mu_m \right) \quad (17)$$

- $m \in \{\text{nat}, \text{wealth}, \text{cog}\}$
- $k_m \text{ } m$
- $w_i^{(m)} \in [0, 1] \text{ } \sum_i w_i^{(m)} = 1$
- $\psi_i^{(m)} : \mathbb{R} \rightarrow \mathbb{R}$ log, probit, z-score
- μ_m, σ_m

where the parameters follow the specifications above.

Table 6: Summary of Feature Transformation Functions

m	i	$\psi_i^{(m)}$	Rationale
nat	GDP/cap	$\psi(x) = \log(x)$	+
nat	Gini	$\psi(x) = \log(1 - x)$	
nat		$\psi(x) = \log(x)$	
wealth		$\psi(x) = \Phi^{-1}(\hat{F}_N(x))$	$\rightarrow$
wealth		$\psi(x) = \Phi^{-1}(\hat{F}_I(x))$	$\rightarrow$
cog		$\psi(x) = \text{z-score}(x)$	-
cog		$\psi(x) = \text{RCR}(x)$	
cog		$\psi(x) = \text{z-score}(x)$	

6.2 Correlation Structure Among Operators

The three potential operators exhibit nonzero but imperfect correlations, with important social-scientific implications:

$$\mathbf{R} = \begin{pmatrix} 1 & \rho_{\text{nw}} & \rho_{\text{nc}} \\ \rho_{\text{nw}} & 1 & \rho_{\text{wc}} \\ \rho_{\text{nc}} & \rho_{\text{wc}} & 1 \end{pmatrix} \quad (18)$$

- $\rho_{\text{nw}} \approx 0.45\text{—}0.65$
- $\rho_{\text{nc}} \approx 0.55\text{—}0.75$
- $\rho_{\text{wc}} \approx 0.30\text{—}0.50$

SCX

These correlations are not part of the definition but empirical features of the social system. They provide input for inequality interaction analysis in the SCX framework.

6.3 SCX Connection to SCX Audit Framework

$\hat{S}$ SCX

The relationship between the social potential operator $\hat{S}$ and the core SCX audit framework is as follows:

- (1) **Input Role** $\hat{S}(e)$ SCX $\hat{S}(e)$ $\hat{S}(e)$ serves as input features for the social-inference module, akin to feature vectors in ML. But when SCX's Audit Sword points at social data, $\hat{S}(e)$ itself can become an audit target.
- (2) **Cercis Decomposition into Cercis Score** Cercis

$$S_{\text{social}}(e) = \underbrace{\alpha^\top \mathbf{S}(e)}_{\text{Potential}} + \underbrace{\eta \cdot N(e)}_{\text{Noise}} \quad (19)$$

$\mathbf{S}(e) = (\hat{S}_{\text{nat}}(e), \hat{S}_{\text{wealth}}(e), \hat{S}_{\text{cog}}(e))^\top \alpha \in \mathbb{R}^3$ The Cercis Score in social data decomposes into a potential-driven component and a noise component.

- (3) **Potential Transfer and Social Mobility** $\Delta S = \hat{S}(e_1) - \hat{S}(e_2)$ — The social potential difference $\Delta S = \hat{S}(e_1) - \hat{S}(e_2)$ measures the space of potential transfer — how much potential could be transferred from high-potential to low-potential entities via social policy. This perspective maps social mobility to a transport problem in the potential field.

7 Deep Structure of Gauge Theory

7.1 Social Potential Field as a Gauge Theory

$\hat{\mathcal{S}}$ $\mathcal{U}$ base manifold $\mathcal{F}(e) = \hat{\mathcal{S}}(e)$ $\mathcal{U}$ $\mathcal{G} = (\mathbb{R}, +)$ $\mathcal{F}$

Social potential $\hat{\mathcal{S}}$ can be formalized as a ***gauge field***. Let $\mathcal{U}$ be the base manifold of social entities, and $\mathcal{F}(e) = \hat{\mathcal{S}}(e)$ a scalar field over $\mathcal{U}$. The gauge group is the additive group $\mathcal{G} = (\mathbb{R}, +)$, acting on $\mathcal{F}$:

$$\mathcal{F}(e) \mapsto \mathcal{F}(e) + c, \quad c \in \mathbb{R} \quad (20)$$

This transformation does not change any physical (social) observable, since only potential differences have operational meaning.

$$\sum_e g_e(\hat{\mathcal{S}}) = 0$$

Group-theoretic interpretation of the gauge-fixing condition:

1. affine space $\mathcal{G}$ Ungauged potential values form an affine space on which $\mathcal{G}$ acts freely.
2. section $\sigma : \mathcal{U} \rightarrow \mathbb{R}$ $\sum_e \sigma(e) = 0$ Gauge fixing is equivalent to choosing a section $\sigma : \mathcal{U} \rightarrow \mathbb{R}$ satisfying $\sum_e \sigma(e) = 0$. This section uniquely specifies the "zero level" of potential.
3. $\mathcal{G}$ Transforms between different sections are global translations, belonging to the action of $\mathcal{G}$.
4. $\mathcal{S} = \{\mathcal{F} : \mathcal{U} \rightarrow \mathbb{R}\} / \mathcal{G}$

All sections form a quotient space $\mathcal{S} = \{\mathcal{F} : \mathcal{U} \rightarrow \mathbb{R}\} / \mathcal{G}$, whose elements are gauge equivalence classes. Gauge fixing selects one representative from each equivalence class.

7.2 Gauge-Invariant Observables

In gauge theory, only gauge-invariant quantities have physical meaning. The following are gauge-invariant:

- (1) **Potential Difference** $\Delta S_{ij} = \hat{\mathcal{S}}(e_i) - \hat{\mathcal{S}}(e_j)$ c $\Delta S_{ij} \mapsto (\hat{\mathcal{S}}(e_i) + c) - (\hat{\mathcal{S}}(e_j) + c) = \Delta S_{ij}$
- (2) **Variance of Potential** $\text{Var}[\hat{\mathcal{S}}] = \frac{1}{N} \sum_e [\hat{\mathcal{S}}(e)]^2 - \left(\sum_e \hat{\mathcal{S}}(e) \right)^2 / N = 0$
- (3) **Rank Correlation of Potentials** $\tau(\hat{\mathcal{S}}_m, \hat{\mathcal{S}}_{m'})$ Kendall τ
- (4) **Potential Gradient** $\nabla \hat{\mathcal{S}} = (\partial \hat{\mathcal{S}} / \partial \text{feature}_i)$

7.3 Alternative Gauge-Fixing Schemes

z-score

While this paper adopts global z-score normalization as the default gauge-fixing scheme, alternative schemes can be considered:

- 1: **Median Gauge** $\hat{\mathcal{S}}'(e) = \hat{\mathcal{S}}(e) - \text{median}_{e' \in \mathcal{U}} \hat{\mathcal{S}}(e')$

More robust to outliers, but sacrifices simplicity of variance interpretation.

- 2: **Min-Gauge** $\hat{\mathcal{S}}'(e) = \hat{\mathcal{S}}(e) - \min_{e' \in \mathcal{U}} \hat{\mathcal{S}}(e')$

Makes all potential values nonnegative; the lowest entity is the baseline zero.

3: Weighted Gauge $\sum_e w(e) \cdot \hat{S}(e) = 0$ $w(e) \geq 0$

Useful for national potential: population-weighted global mean zero (rather than country-count mean).

4: Anchor-Point Gauge $\hat{S}(\text{anchor}) = 0$ Set potential of one or more "reference entities" (e.g., OECD average) to zero. Convenient for policy analysis and intertemporal comparison.

Remark 7.1.

The choice of gauge-fixing scheme is itself a meta-level social-choice problem. Different schemes do not change any gauge-invariant quantities (potential differences, rank correlations), but they affect visualization interpretation and policy-narrative framing. Therefore, applications should explicitly state the scheme used and provide sensitivity analysis with at least two schemes.

8 Extensions and Future Work

8.1 Multi-Resolution Gauge Fixing

Current definitions use global gauge fixing. But in large heterogeneous populations, global gauges may mask local structure. Multi-resolution gauge fixing partitions the population by hierarchy (geographic, administrative, cultural) and fixes gauges within each subgroup:

$$\hat{S}^{(L)}(e) = \frac{\tilde{S}(e) - \mu_h(e)}{\sigma_h(e)}, \quad h(e) \in L \quad (21)$$

8.2 Spring Dynamic Potential and Spring Updates

SCXSpring

In SCX's Spring framework, social potential evolves dynamically with data updates. Define the time derivative of potential:

$$\dot{\hat{S}}_m(e, t) = \frac{\partial \hat{S}_m(e, t)}{\partial t} \approx \frac{\hat{S}_m(e, t + \Delta t) - \hat{S}_m(e, t)}{\Delta t} \quad (22)$$

$\dot{\hat{S}} > 0$ $\dot{\hat{S}} < 0$ — Potential Flux

$\dot{\hat{S}} > 0$ indicates rising potential (upward social mobility), $\dot{\hat{S}} < 0$ indicates declining potential. The rate of potential change constitutes a new social indicator — **potential flux** — whose distributional characteristics reflect the speed and direction of social mobility.

8.3 Adversarially Robust Audit of Potential Operators

SCXAR-Theorem

Quantification of social potential inevitably faces potential gaming from various stakeholders: nations may underreport Gini coefficients, individuals may inflate income, researchers may manipulate citation metrics. This directly connects to SCX's AR-Theorem.

AR

$$\hat{S}^{\text{AR}}(e) = \inf_{\delta \in \mathcal{B}(e)} \hat{S}(e; x_1 + \delta_1, \dots, x_k + \delta_k) \quad (23)$$

$\mathcal{B}(e)$ e feasible manipulation ball

The AR-augmented potential operator takes the infimum over feasible manipulation balls, ensuring an adversarially robust lower bound on potential values.

9 Conclusion

SCX $\hat{S}$ —————

This paper defines the complete operational form of the potential operator $\hat{S}$ for the social-inference module of the SCX egalitarian framework. Three dimensions — national potential, wealth potential, and cognitive potential — cover key aspects of social inequality, and all satisfy four axiomatic conditions: quantifiability, auditability, gauge fixing, and cross-domain comparability.

Summary of Core Contributions:

(1) -17

Unified form: all three operators share a unified aggregation-normalization architecture (Eq. 17), making potential values directly comparable across dimensions.

(2) z-score $\sum_e \hat{S}(e) = 0$

Gauge fixing: z-score normalization automatically satisfies $\sum_e \hat{S}(e) = 0$, with discussion of alternative schemes and their social-choice implications.

(3) 1

Audit protocols: each operator is accompanied by a detailed audit protocol (Algorithm 1), ensuring computations can be reproduced and verified by any independent auditor.

(4) / logprobitRCR *Feature engineering: psychologically/economically justified transformation functions (log, probit, RCR) for each dimension, adapting raw data to normality assumptions.*

(5) SCX *Gauge-theoretic connection: formalized social potential as a gauge field theory, establishing a bridge between the field-theoretic language of social inequality and the SCX audit framework.*

Next Steps:

1. WDI, SWIID, CHFS, CFPS, OpenAlex
2. Python/R
- 3.
4. SCXSprint
5. Q-Theorem

Empirical calibration with real-world data; open-source reference implementation; sensitivity analysis of gauge-fixing schemes; integration into SCX Sprint active-learning loop; quantum generalization for privacy-preserving aggregation.

Table 7: Notation Table

Symbol	Meaning
$\mathcal{U}$	Universe of social entities
$\hat{\mathcal{S}}(e; m)$	m e Potential of entity e under dimension m
$\tilde{\mathcal{S}}_m(e)$	Raw (ungauged) potential
μ_m, σ_m	m Global mean and std of dimension m
$w_i^{(m)}$	m i Weight of feature i in dimension m
$\psi_i^{(m)}$	m i Normalization transform for feature i
Φ^{-1}	Quantile function of standard normal
$x_1^{(m)}$	m First observable feature in dimension m
$\hat{\mathcal{S}}_{\text{nat}}$	National potential operator
$\hat{\mathcal{S}}_{\text{wealth}}$	Wealth potential operator
$\hat{\mathcal{S}}_{\text{cog}}$	Cognitive potential operator
$\mathcal{G}$	$\mathbb{R}$ Gauge group (additive group $\mathbb{R}$)
RCR	Relative Citation Rate
$\mathcal{A}_{\text{nat}}$	National potential audit protocol

A Appendix: Notation Table

B JSON Appendix: JSON Schema for Audit Trail

```
{
  "entity_id": "CHN",
  "dimension": "national",
  "timestamp": "2026-07-01T00:00:00Z",
  "raw_features": {
    "gdp_per_capita_ppp": 21500.0,
    "gini_coefficient": 0.382,
    "life_expectancy": 78.2
  },
  "feature_sources": {
    "gdp_per_capita_ppp": {
      "source": "World Bank WDI",
      "indicator": "NY.GDP.PCAP.PP.KD",
      "year": 2023,
      "api_version": "v2"
    },
    "gini_coefficient": {
      "source": "SWIID",
      "version": "9.4",
      "estimation_type": "direct_survey",
      "year": 2022
    },
    "life_expectancy": {
      "source": "WHO GH0",
      "indicator": "WHOSIS_000001",
      "year": 2023
    }
  }
},
```

```

"transformed_features": {
  "log_gdp": 9.9758,
  "log_complement_gini": -0.4780,
  "log_life_expectancy": 4.3593
},
"weights": {"gdp": 0.333, "gini": 0.333, "life_exp": 0.333},
"raw_potential": 3.2857,
"gauge_parameters": {
  "mu": 3.1192,
  "sigma": 0.5438,
  "reference_population_size": 195,
  "gauge_scheme": "z-score"
},
"gauge_fixed_potential": 0.306,
"audit_hashes": {
  "sha256_input": "abc123...",
  "sha256_output": "def456..."
},
"warnings": []
}

```

C Python Appendix: Python Reference Implementation Skeleton

```

import numpy as np
from scipy import stats
from typing import Dict, List, Optional, Tuple

class PotentialOperator:
    """SCX Social Potential Operator S-hat.

    Implements the gauge-fixed potential operator for SCX social inference.
    Supports national, wealth, and cognitive dimensions.
    """

    def __init__(self, dimension: str, weights: Optional[Dict[str, float]] = None):
        self.dimension = dimension
        self.weights = weights or self._default_weights()
        self.mu: float = 0.0
        self.sigma: float = 1.0

    def _default_weights(self) -> Dict[str, float]:
        defaults = {
            "national": {"gdp": 1/3, "gini": 1/3, "life_exp": 1/3},
            "wealth": {"net_asset": 0.5, "income": 0.5},
            "cognitive": {"education": 1/3, "citation": 1/3, "reasoning": 1/3},
        }
        return defaults.get(self.dimension, {})

    def transform_features(self, raw: Dict[str, float]) -> Dict[str, float]:

```

```

    """Apply dimension-specific feature transforms."""
    ...

def aggregate(self, transformed: Dict[str, float]) -> float:
    """Weighted sum of transformed features."""
    return sum(
        self.weights[k] * transformed[k]
        for k in self.weights if k in transformed
    )

def fit_gauge(self, raw_potentials: np.ndarray) -> None:
    """Compute global mu and sigma for gauge fixing."""
    self.mu = np.mean(raw_potentials)
    self.sigma = np.std(raw_potentials, ddof=0)

def gauge_fix(self, raw_potential: float) -> float:
    """Apply z-score gauge fixing."""
    return (raw_potential - self.mu) / self.sigma

def compute(self, entity_features: Dict[str, float]) -> float:
    """Full pipeline: transform -> aggregate -> gauge-fix."""
    transformed = self.transform_features(entity_features)
    raw = self.aggregate(transformed)
    return self.gauge_fix(raw)

def verify_gauge(self, potentials: np.ndarray, tol: float = 1e-10) -> bool:
    """Verify sum of gauge-fixed potentials is zero."""
    return abs(np.sum(potentials)) < tol

# Example usage:
op_national = PotentialOperator("national")
features = {"gdp": 65000, "gini": 0.30, "life_exp": 83.5}
S_nat = op_national.compute(features)
print(f"National potential: {S_nat:.3f}")

```

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SCX SCX This document is the formal technical specification of SCX social-inference operator definitions. All operator definitions carry audit protocols, satisfying the verifiability requirements of SCX's Audit Sword.